

Phase 4 — Model Evaluation and Explainability

Business Goal

Ensure that predictions are reliable, interpretable, and safe for hospital operations and finance teams.

- Evaluate both models on training and test data using confusion matrices and precision, recall, and F1-score.
- Compute business metrics such as recall for High Risk class and recall for Rejected claim class.
- Generate feature importance and explainability summaries.
- Segment performance by gender, city, and insurance_provider to identify fairness gaps.
- Create a consolidated model card documenting performance, limitations, and assumptions.

Deliverables

- Risk model evaluation report
- Claim model evaluation report
- Model card document
- Explainability summary

Recommendation (To handle imbalance data)

1. **Tune Hyperparameters:** Adjust class weights or tree depth.
2. **Feature Engineering:** Create more interaction features (e.g.,
 - bill_amount
 - vs
 - department
 - averages).
3. **Resampling:** Use SMOTE or undersampling to handle the class imbalance more aggressively.